MEMORANDUM

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Date: 15 October 2025

Subject: Hypothesis Testing with Multiple OLS Linear Regression of Spotify Track Popularity

The year 2020 marked dramatic changes in music consumption, driven by the global COVID-19 pandemic and related lockdowns. People spent more time at home and live music events disappeared, making Spotify a key platform for discovering, enjoying, and coping with music. Analyzing how song features such as danceability, energy, genre, and musical mode influenced track popularity during this unique year provides important insights into shifting listener behaviors and emerging trends in digital music during the pandemic.

The target population for this analysis includes all tracks released on Spotify in 2020 within the genres of rock, EDM, hip-hop, or acoustic. The findings represent the popularity factors of these songs during the start of the COVID-19 pandemic. The hypothesis is that Spotify tracks released in 2020 with higher danceability, greater energy, major mode, and specific genres will be associated with increased popularity. This analysis found that tracks with higher danceability and those in the hip-hop genre achieved significantly higher popularity. In contrast, tracks in the EDM and acoustic genres achieved significantly lower popularity compared with rock, while energy and major mode showed no significant effects on popularity.

The dataset for this analysis includes the Spotify 1 Million Tracks dataset (Joshi et al., 2023). This analysis contained 2398 observations focused on four key genres of 2020: acoustic, hip-hop, EDM, and rock. The dependent variable is track popularity, measured as a numeric score from 0 to 100 reflecting Spotify user engagement. The independent variables include genre, danceability and energy (both continuous measures ranging from 0.0 to 1.0), and mode (binary: 1 = major key, 0 = minor key). There are no missing values in any of the variables analyzed. Three tracks were identified as outliers in the energy variable; however, their presence did not influence the overall findings. All tests were conducted using an alpha level of 0.01.

Table 1 shows that acoustic and hip-hop genres dominate the sample, comprising 76% of tracks, while EDM and rock represent smaller portions of the data. Major key compositions (Mode) account for nearly two-thirds of all songs, with minor key tracks making up the remainder. Within the 4 chosen genres, Spotify's catalog in 2020 reflected a strong preference for acoustic and hip-hop

music, and that most tracks were written in major rather than minor keys, aligning with trends towards uplifting and rhythm-driven content in this unique year (within these genres).

The univariate analysis of numerical variables (Table 2) shows that song popularity in the Spotify subset ranges from 12 to 84, with both the mean of 43 and median of 46 being close to each other, indicating most tracks in the 4 genres have below-average to moderate popularity. Table 2 illustrates that danceability and energy scores for tracks in the 2020 Spotify subset cluster around moderate-to-high values, with means of 0.64 and 0.54, and very little skewness or spread. This indicates that, among songs released and present in this dataset, most have a balance of rhythmic engagement and energetic qualities. These features reflect the types of tracks that were most commonly produced, released, and included on the platform in that year.

Table 1: Categorical Summary

Genre	Freq.	Percent
Acoustic	951	39.7%
Hip-Hop	854	35.6%
EDM	446	18.6%
Rock	147	6.1%
Total	2398	100.0%
Mode	**Freq.**	**Percent**
Major key	1560	65.1%
Minor Key	838	34.9%
Total	2398	100.0%

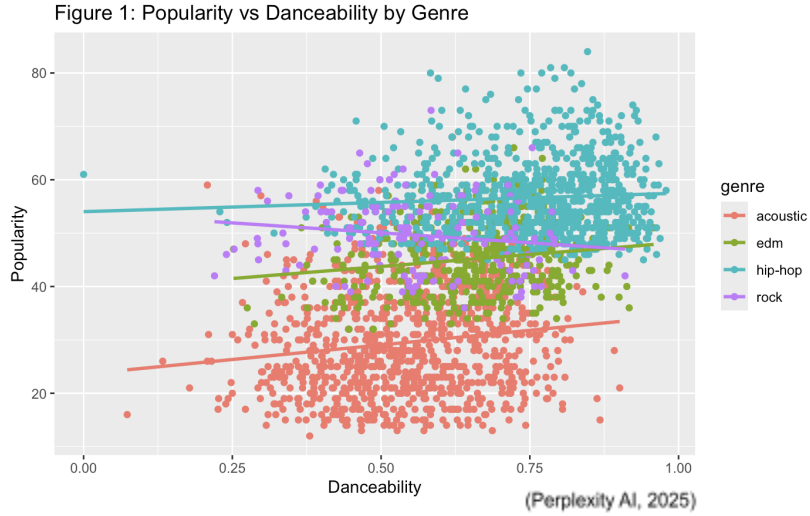
Table 2: Quantitative Summary

	Freq.	Min.	Median	Mean	Max.	St. Dev	Skewness
Popularity	2,398	12.0	46.0	43.3	84.0	14.5	-0.21
Energy	2,398	0.01	0.57	0.54	0.99	0.25	-0.29
Danceability	2,398	0.00	0.64	0.64	0.98	0.16	-0.20

Regression Formula: Popularity = 46.35 − 5.16(EDM) + 5.67(Hip-hop) − 19.99(Acoustic) − 0.17(Energy) + 6.60(Danceability) − 0.45(Mode)

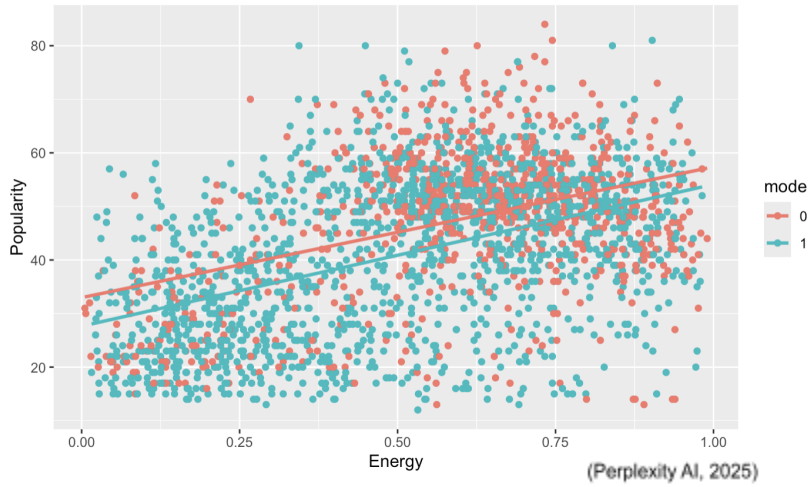
The regression analysis (Table 3) reveals several key insights into the factors associated with track popularity in the dataset. Danceability emerges as a significant positive predictor, with higher danceability scores linked to greater popularity. This relationship is visually supported by Figure 1, which shows a clear upward trend for all genres besides rock, which displays the opposite relationship. Hip-hop songs exhibit the highest popularity, whereas acoustic tracks exhibit the lowest despite it

Figure 1: Popularity vs Danceability by Genre



making up the highest percentage of observations. Energy, by contrast to danceability, does not appear as a significant predictor of popularity, and both the scatterplot in Figure 2 and results in Table 3 show substantial overlap and no meaningful difference between major (mode 1) and minor (mode 0) tracks. The regression lines in Figure 2 are nearly parallel, reinforcing the result that mode is not a substantial driver of popularity in this sample. Table 3 and Figure 1 and 2 show that genre and danceability are the strongest predictors of track popularity (hip-hop has a positive effect, while acoustic and EDM are negative) whereas energy and musical mode do not significantly affect popularity. The overall regression model is highly statistically significant, with an Adjusted R^2 of 0.675, indicating that 68% of the variance in track popularity is explained by the included variables (Table 3). The model has substantial explanatory power (close to 1), as confirmed by a large F-statistic (832.1), and a Cohen's f^2 of 2.08, which is also very large.

Figure 2: Popularity vs Energy by Mode



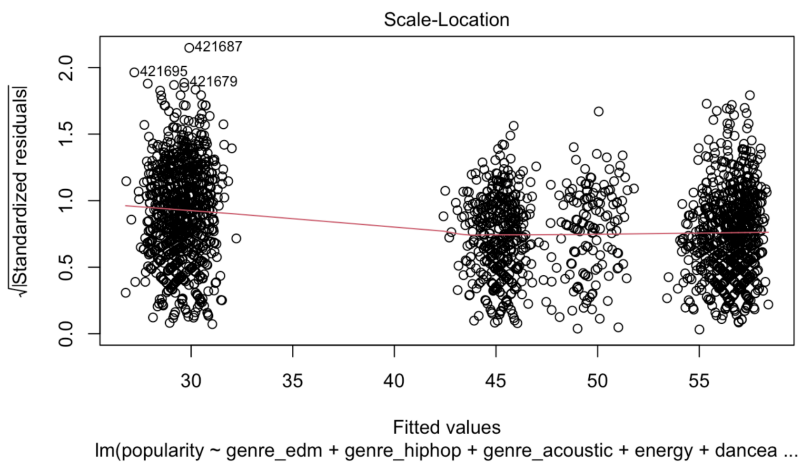
One challenge with this analysis is the choice of genres, since leaving out pop may have made it harder to see more interesting patterns in other variables like mode, energy, or danceability. A more balanced genre selection or finding a way to include pop (without overwhelming the analysis) might have helped reveal new trends or relationships. It could be useful to use pop music as DV, which might show different results for what made a song popular during 2020. Additionally, there are minor concerns in the diagnostic plots: the scale-location plot shows a slight "bounce" pattern, suggesting small heteroscedasticity, while the Q-Q plot curves a bit at the ends, indicating mild non-normality in the tails. These issues are unlikely to strongly affect results, but they could be reduced by addressing outliers or double-checking for unmodeled variables in future analyses.

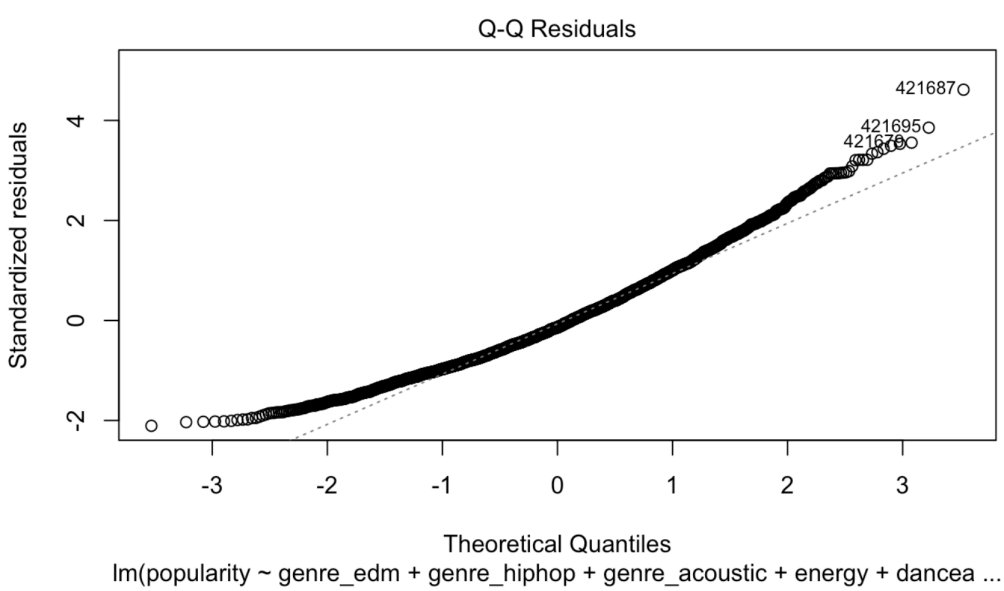
This analysis shows that, for 2020 Spotify tracks in the chosen genres, songs that are more danceable and hip-hop influenced tend to be more popular, while energy and mode don’t matter as much. Based on these results, consulting teams should advise music clients to focus on producing and promoting tracks that have strong danceable rhythms and hip-hop features to boost streaming success. Using data on what makes songs popular can help guide smarter decisions for releases and song recommendations.

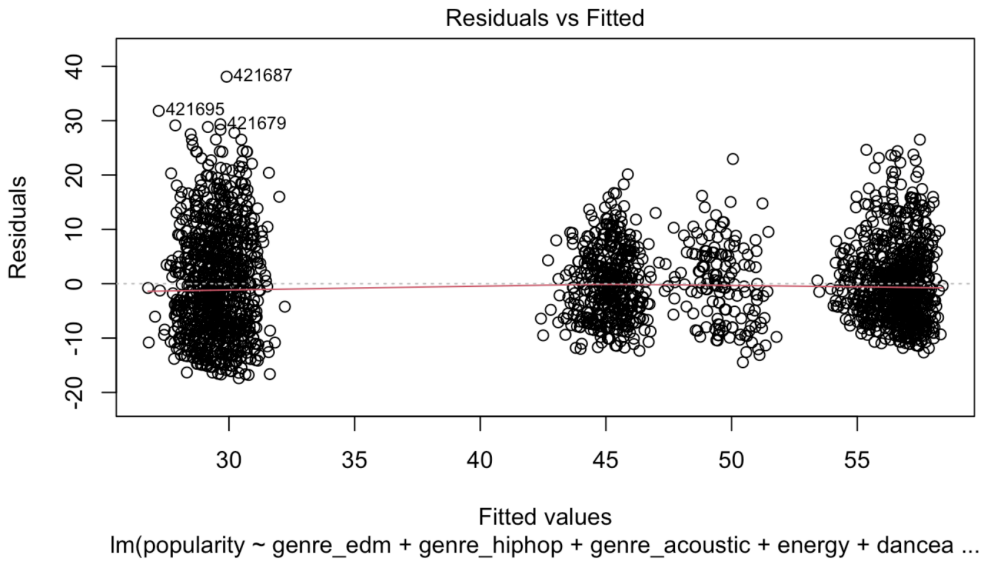
Additional Tables/Plots

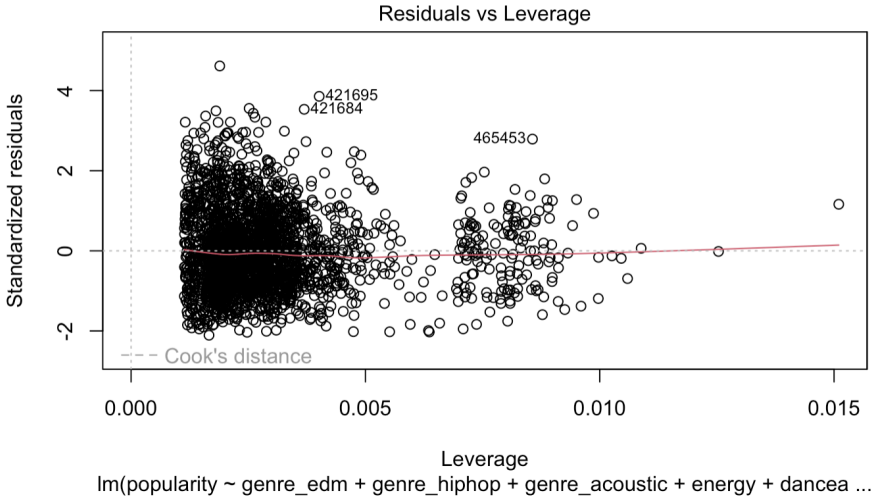
Table 3: Regression Results

	Dependent variable:
	popularity
genre_edm	-5.158***
	(0.802)
genre_hiphop	5.673***
	(0.780)
genre_acoustic	-19.994***
	(0.799)
energy	-0.168
	(0.959)
danceability	6.600***
	(1.287)
mode	-0.451
	(0.367)
Constant	46.352***
	(1.209)
Observations	2,398
R2	0.676
Adjusted R2	0.675
Residual Std. Error	8.262 (df = 2391)
F Statistic	832.124*** (df = 6; 2391)
Note:	*p<0.1; **p<0.05; ***p<0.01






Residuals vs Fitted
Residuals
Fitted values
lm(popularity ~ genre_edm + genre_hiphop + genre_acoustic + energy + dancea ...
421687
421695
421679
40
30
20
10
0
-10
-20
30
35
40
45
50
55


Residuals vs Leverage
Standardized residuals
Leverage
lm(popularity ~ genre_edm + genre_hiphop + genre_acoustic + energy + dancea ...
421695
421684
465453
Cook's distance
4
2
0
-2
0.000
0.005
0.010
0.015

References

Joshi, A., Parolkar, A., & Das, V. (2023, June 21). *Spotify_1million_tracks*. Kaggle. https://www.kaggle.com/datasets/amitanshjoshi/spotify-1million-tracks

Perplexity AI. (2025). Perplexity [Large language model]. https://www.perplexity.ai/search/how-can-i-make-a-scatterplot-u-8goTzbvkQFC0ThhTtxSsIw